



PRODUCTION FACILITY & OUTPUT AUDIT REPORT

For PURO.EARTH

Company / CO ₂ Removal Supplier	West Fraser, LLC
Production Facility Name	Joanna Facility
Production Facility Address	564 Woodyard Rd, Kinards, South Carolina, 29355
Production Facility Coordinates	34°22'45"N 081°46'07"W
Net Volume of CO ₂ Removal	5995.27 CORCs
Removal Method	Biochar
Removal Period	07/23-12/23
Auditors	Bill Chatterton Kyann Hadife 350Solutions, Inc.

ISSUED: NOVEMBER 15,
2024



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PRODUCTION FACILITY & OUTPUT AUDIT REPORT

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Removal Method: Biochar		
Report Date: 11-7-2024		
Document No: 350VP-WF-PU2406		
Rev: 2.4		

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1. INTRODUCTION

350Solutions was contracted to perform an audit and validation of the production facility as well as verification of carbon dioxide removal credit (CORC) on West Fraser's (WF) Joanna Mill in Kinards, South Carolina. The facility produces Oriented Strand Board (OSB) and generates syngas by gasifying waste biomass to produce heat for steam production essential for OSB manufacturing. This process results in the continual production of biochar. To address biochar accumulation, West Fraser has partnered with a local waste management company and soil additive manufacturer to utilize the biochar in soil products.

350Solutions conducted an audit of the process, lifecycle CO₂ emissions assessment (LCA), and other administrative details to verify compliance with the requirements of the Puro.Earth Standard General Rules v3.1 [1] and Biochar Methodology (Edition 2022 Version 3) [2]. The audit and verification began with a document review and followed with a site visit and detailed data audit on July 11th, 2024. Additional files were requested and reviewed by 350Solutions to clarify the findings of the initial investigation.

Moreover, Puro.earth is currently transitioning to use the 4.0 version of the Puro General Rules [3]. Although the West Fraser facility and output audit described in this report were conducted using version 3.1 of the General Rules, certain rules described in the updated version of the document (v4.0), such as the Ongoing Issuance Rights, are applicable to all projects currently registered or in the process of getting registered on the registry. Section 5 of this report aims to assess whether the Joanna Mill meets the Ongoing Issuance Rights requirements outlines in Appendix A of the updated Puro General Rules (v4.0)

Based on the activities above, 350Solutions issued an initial version of this report on 09/12/2024 highlighting non-compliance areas needing to be addressed by West Fraser to be eligible for CORC issuance under the Puro registry, following the guidance of the Puro Biochar Methodology (v3.0). Between 09/12/2024 and 10/04/2024, West Fraser has provided files to address the concerns raised during the audit period. As a result of the amendments made, the total volume of CORCs verified by 350Solutions was 5995 as opposed to the 6022 initially claimed by the supplier. A summary of the issues raised by 350Solutions, and mitigating actions taken by West Fraser is presented in Appendix 1: Log of previous audit issues and responses of this report.

TABLE 1. WEST FRASER VERIFICATION SUMMARY

Verification Summary	
CO₂ Removal Supplier	West Fraser
Production Facility Name	Joanna
Facility ID	320449
Removal Method	Biochar – Production from woody forest waste and industrial process waste and use as soil amendment
Claimed CORC Factor	2.17 t CO _{2e} / t biochar
Verified CORC Factor	2.16 t CO _{2e} / t biochar
Claimed CORCs	6022 CORCs
Verified CORCS	5995.27 CORCs
Verification Report Date	10-15-2024
Site Visit Date	07-11-2024
Verification Period	07-01-2023 to 12-31-2023
Production Facility Location (Address and GPS Coordinates)	564 Woodyard Rd, Kinards, South Carolina 34°22'45"N 081°46'07"W
Verification Type	Combined Production Facility Audit and Output Audit for Puro.Earth, including on-site visit and facility audit; Puro Standard General Rules (v3.1) and Annex A: Biochar Methodology (v3)

2. TECHNOLOGY DESCRIPTION

WF has developed a slow pyrolysis system that uses woody biomass to generate heat for steam and electricity production. The process also produces biochar used for soil amendment. The biochar, a charcoal-like substance with high carbon content, is made from woody biomass at WF's Joanna facility, which produces Oriented Strand Board (OSB).

The facility utilizes gasification of waste biomass—fuel chips from non-merchantable wood, bark, sawdust, shavings, and hog fuel—to generate syngas. The biomass feedstock at the Joanna facility comes from three sources:

- Purchased utility-size waste wood chips and grindings from harvested treetops and limbs, including Southern Yellow pine family species and various hardwoods. The chipping and grinding occur in the forest.
- Purchased waste wood, including bark, sawdust, and hog fuel from sister mills and forest thinning to support forest health. These materials arrive as moist chips.
- Waste bark from log peeling in the OSB board production at the Joanna facility.
- OSB boards unfit for sale produced at the Joanna facility.

Biomass waste from sister mills is trucked into Joanna and piled up onsite (see images in PDD). This waste along with rejected OSB boards is moved onto conveyors that bring the biomass into the kilns, by diesel diggers. Biomass from Joanna's own log peeling is collected on site and moved onto conveyors that bring it to the kilns.

Purchased wood chips and grindings processed offsite from Sustainable Forestry Initiative (SFI)-audited¹ landscapes, trucked into Joanna, piled onsite (see PDD images) and moved onto conveyors that bring the biomass into the kilns, by diesel diggers.

The energy system at the Joanna facility consists of three parallel systems (A-Train, B-Train, and C-Train) that convert feedstock to outputs using wood waste, with natural gas as an auxiliary fuel. Each system includes a rotary kiln gasifier, a secondary combustion chamber, a saturated steam boiler, a multicyclone, and an induced draft fan. C-Train also has a thermal oil heater. The exhaust from these systems is directed to an electrostatic precipitator (ESP) and then to a common stack. Syngas generated from the gasifiers is combusted, and the resulting heat is used in the steam boilers and thermal oil heater.

The gasification process continuously produces biochar, which has been accumulating on-site at West Fraser for over 24 years (Figure 1) . With evolving permitting requirements, the company is prohibited from stockpiling biochar on its site. It is now seeking a more efficient way to manage this 'waste' byproduct from the OSB production process. The most cost-effective options are to either burn the biochar on-site or explore profitable offtake agreements for its use elsewhere. However, the potential to generate revenue through carbon credits, along with West Fraser's commitment to environmental stewardship, has driven them to seek alternative applications for biochar. As a result, they have established a partnership with a local waste management company and a soil additive manufacturer to incorporate the biochar into various soil products.



FIGURE 1. HISTORIC BIOCHAR PILE ACCUMULATED AT THE WEST FRASER JOANNA MILL

Biochar outfeed from the three streams is collected into one pile and transported to a new storage area managed by Southeast Organics (SEO) (Figure 2).

¹ This was not verified as part of this audit. West Fraser is an SFI certified. All accreditation records can be found at: [Forest Certification | West Fraser](#)

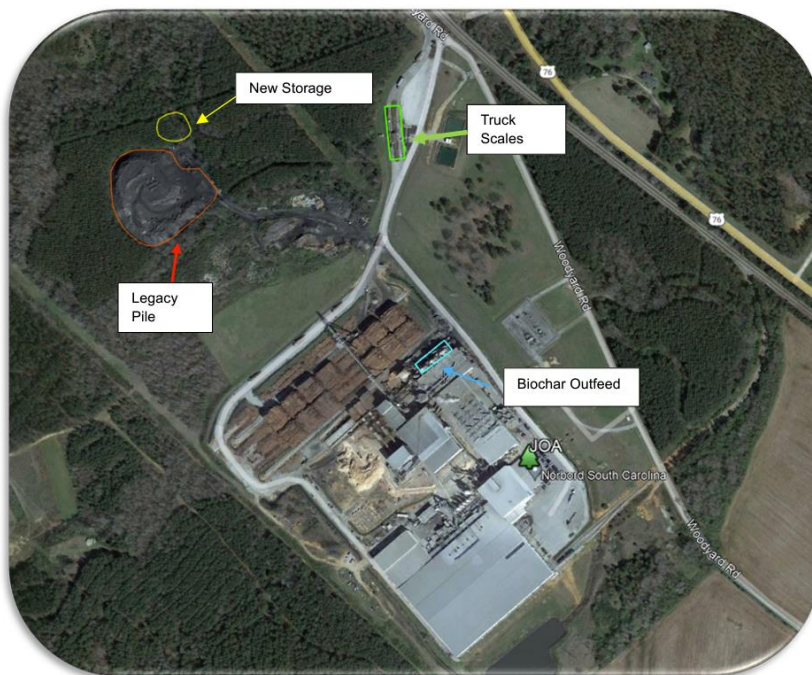


FIGURE 2. ARIAL VIEW OF THE JOANNA FACILITY

At the new storage area, biochar is screened for rocks and wood chips and processed by SEO before being transported to Scotts Hyponex, which uses the biochar as an additive in soil products sold in retail stores within a 100-mile range of Chester, South Carolina. The chain of custody of the biomass is summarized by Figure 3.

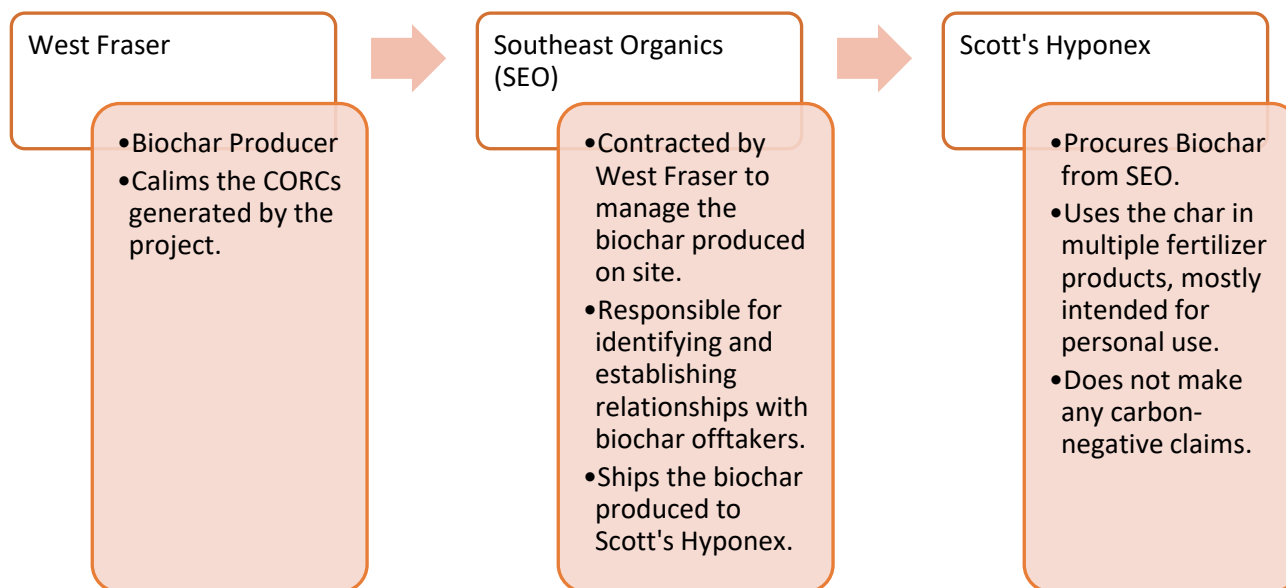


FIGURE 3. DIAGRAM OF THE BIOCHAR CHAIN OF CUSTODY

The biochar production process and sub-processes used by WF for carbon removal are outlined in accordance with the Cradle-to-Grave Life Cycle Assessment (LCA) system boundaries (Figure 4) of the Puro Standard General Rules. It is important to note that the secondary natural gas combustion chamber in the diagram is used to a) manage syngas emissions generated by the kiln during pyrolysis and b) generate heat for steam production by combusting natural gas. As per the Puro Biochar Methodology (V3), “Operation of the pyrolysis reactor and post-pyrolysis equipment (e.g. combustion chamber for pyrolysis gases and oil, flue gas treatment systems) or operation of the gasifier reactor and post-processing equipment”. This implies that only the emissions generated by syngas processing in the secondary chamber are allocated to the biochar. Natural Gas combustion is outside of the system boundary of the project (i.e. the part of the mill used to produce biochar).

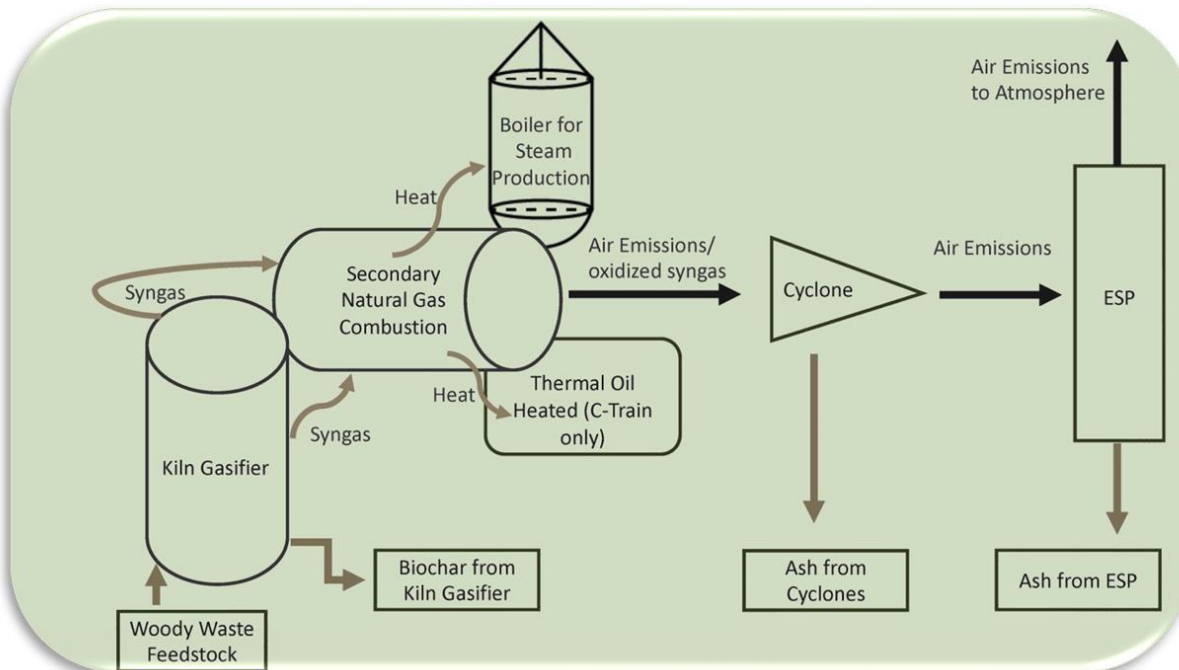


FIGURE 4. SYSTEM BOUNDARY AND PROCESS FLOW DIAGRAM FOR THE WEST FRASER BIOCHAR PROCESS

The following operations at the Joanna facility were observed and assessed during verification and site audit:

- Receipt of biomass from wildfire reduction treatment harvests, with documentation of feedstock delivered quantity, and acceptance of sawmill waste wood;
- Woody biomass from the forest is chipped and mixed with sawmill waste wood, and conveyed to the pyrolysis process;
- Operation of the pyrolysis reactors, including:
 - Pyrolizer input (average 18.73 tonne wet biomass input/load biochar);
 - Production and combustion of syngas;
 - Recovery of and cycling of combustion gas heat for steam generation and pyrolysis;
 - Collection, cooling of biochar;
- Storage and packaging of biochar (SEO);
- Sale and shipping of biochar to Scotts Hyponex;
- Documenting all process operations, including input and output rates, biochar production and product sales, and amounts of by-products.
- Sampling and analysis of the biochar product.

3. AUDIT SUMMARY

3.1 AUDIT APPROACH

A planned series of audit activities were conducted by 350Solutions to independently validate and verify the production facility, its operations, production, and output data, and CORC claims. The audit was conducted following the specifications of Puro General Rules (v3.1) and the Biochar Methodology (v3). Specific audit activities conducted are summarized in Table 2. Audit Activities. A completed Puro Biochar Methodology Compliance Checklist used during the verification is attached to this report as Appendix 2: Puro.Earth Biochar Methodology Audit Checklist. Verifier qualifications are attached as Appendix 4: Verifier Qualifications.

TABLE 2. AUDIT ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed*
06/19-26/2024	Introductory Document Review	- Review of LCA - Review of Puro CORC calculations - Review of facility registries and permits	- Puro_Registration_320449.pdf - Tab 04.09 Certificate of Merger.pdf - West Fraser EWP LLC certificate of formation filed with DE SOS 11-27-2023 - CORC Report Summary - Biochar_West Fraser.xlsx - LCA calculation West Fraser February 2024 v3.xlsx - Air permit.pdf - BOW_IndustrialStormwaterGP2022 - JOA_Environmental Evaluation Report_v1_signed.pdf - BB_Stakeholder Engagement Report_ver2.docx
06/26/2024 – 07/10/2024	Data Review	- Review of waste biomass sources and sustainability - Review of system inputs and outputs - Review evidence of product output - Review of biochar properties - Review of biochar end use	- Biochar Financial Model- 05.12.23.xlsx - Control Labs 10-23-2023 Analysis.pdf - Final PDD West Fraser- Joanna Facility. 3.5.2024 - Ash moisture readings.docx - JOA_Terms and Conditions.pdf - NSF CoC C0750561-CC1 (1).pdf - NSF Fiber Sourcing C0750561-FS2 (2).pdf - Residual and log purchase contracts at West Fraser.docx - Sustainable Forestry Initiative (SFI) Fibre Sourcing - all wood procurement WF US mill.pdf - Signed Declaration from Scott's.pdf - SEO Agreement_ Section 4.2_.pdf - Biomass types and origins list.xlsx - Joanna Biochar OSHA-WHMIS-GHS SDS 2024-03-01.pdf - Joanna_West Fraser Ash SDS.pdf - Mass and energy balance of production process_Final.xlsx - Biochar production equipment questionnaire_June 2024.xlsx - Puro additionality - West Fraser- Joanna June 2024.docx - Control Labs 10-23-2023 Analysis.pdf - Signed Declaration from Scott's.pdf - Joanna Ash Totals-SEO.xlsx - Soil Temperature Selection Methods.docx - July 2023 Wood Usage.pdf - August 2023 Wood Usage.pdf - Sept 2023 Wood Usage.pdf - Mill Wood Usage Oct 2023.pdf - Mill Wood Usage Nov 2023.pdf - Mill Wood Usage Dec 2023.pdf

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07/11/2024	On-site Visit	- Opening meeting and process walk through - Witness of operations, measurement points, and instrumentation - Review of equipment and calibrations - Review of intake and production data collection - Confirmation of company and facility administrative details - Confirmation of facility environmental and social safeguards	- Verifier observations of operations, measurement points, and instrumentation (see compliance checklist) - Observation and documentation of feedstock receipt, production logs, equipment utilized, data collected, material handling equipment and procedures - Weigh scale calibration (7-28-23 Inbound truck scales certification, calibration tests (1).pdf - 7-28-23 Outbound truck scales certification, calibration tests (1).pdf - 9-27-23 Inbound truck scales certification, calibration tests (1).pdf - 9-27-23 Outbound truck scales certification, calibration tests (1).pdf & Joanna Ash Totals-SEO/xlsx). - On-site review of all documents listed above
07/11/2024 – 10/7/2024	Post-visit Request for Information and review of post-visit files (See Appendix 3: 350Solutions/West Fraser Post-Visit Request for Information)	Follow-up request for information sent to West Fraser on: - Ash Handling practices and records - Biochar characterization and testing - Financial Additionality - Additional process logs Discussion with Accend on: - LCA assumptions and omissions	- Puro_WF_Accend_LCA_Post-visit_RFI_Accend responses 16.07.2024 - West Fraser - SC Joanna GHG Calculation_2023-0918.xlsx - Javelin redacted agreement.docx - #1 - West Fraser - SC Joanna GHG Calculation_2023-0918.xlsx - 9 13, 2022 ESP, Dryer 1-3 Zone 2 Results Summary_REV111822.xlsx - Air permit.pdf - Aug '23-Jan '24 Joanna24 Week 1 Stoplight – January.xlsx - BOW_IndustrialStormwaterGP2022.pdf - Copy of Ash Management (002).xlsx - email explaining 12% roundwood- bark.docx - Scale calibrations.msg - Stormwater permit.msg - exhibita_346757.pdf - Joanna Biochar sampling and dry mass calculation protocol_ver2.docx - Joanna Financial Additionality (2).xlsx - Joanna FUEL-Moisture Content.xlsx - Post-audit RFI - 350Solutions_response 1.xlsx - Updated SEO declaration.pdf - 350Solutions video conferencing call with Accend on 07/17/2024 - Ash Life # 2-ver 2(clean).docx

3.2. PROCESS INPUTS & OUTPUTS

The West Fraser process uses waste biomass as its primary feedstock, with air input to support gasification (internal heating), emission control via oxidizer combustion of syngas, and production of biochar. Energy-related inputs include fuel (diesel) for materials handling equipment, electricity for systems operation, and fuel (waste biomass or natural gas) for pyrolysis startup. The process uses water for biochar handling safety and moisture content management for final product. Operating time during this reporting period is from 07/01/23 to 12/31/23 for a total of 3818 hours² of production. Table 3 summarizes the observed inputs and outputs from the process and typical rates from supplied operational data.

² Operational hours assume 100% uptime. In-practice, plant uptime at the Joanna Mill is around 92-95% according to plant personnel.

Table 3. Verified Production Facility Inputs & Outputs

Input/Output	Verified Rate	Notes (Specifications, source, etc.)
Waste Biomass input (forest and sawmill woody waste)	Average 13.6 tonne wet biomass input/dry tonne biochar	Typically, nominal 50% internal sawmill waste and 50% external wood waste.
Biomass supply inputs (collection, handling, transportation emissions), (Ebiomass)	21 kg CO ₂ -eq. /tonne biochar DMB	Emissions are from the transport of biomass feed to facility. Average roundtrip transport distance is 100 mi.
Production inputs: - diesel (materials handling equipment) - site electricity use	1.34 L/tonne dry biochar 754.69 kWh/tonne dry biochar	Natural gas used for process startup and stabilization, and reactor warming during periods of extremely cold ambient conditions. These values were conservatively estimated in the LCA.
Production water input	Not recorded at the time of the audit	Water use for biochar quench (as needed), is estimated to be equivalent to the moisture content of the biochar. Emissions associated with estimated water use are negligible and not currently included in Eproduction.
Biochar production output	8,157 total tonne (Dry) 2773 total tonne (Dry) Used	Biochar production during normal operations from LCA data and production logs. Kiln operated 3818h from 07/2023 to 12/2023
Production and operation emissions output (Eproduction)	119 kg CO ₂ e/tonne biochar DMB	Emissions from all production process and supporting equipment, including process startup, burnout, process operations, syngas destruction, material handling equipment fuel usage, packaging, and equipment manufacture.
Product distribution emissions output (Euse)	27 kg CO ₂ e/tonne biochar DMB	Verified based on average transport distance from production facility to purchasers, and end use application. Evidenced by truck fuel consumption records.

3.3. VERIFIED OUTPUT & CORCS

Table 4 includes the specific CORCs claimed by West Fraser for its Joanna facility, as well as the level verified by 350Solutions during the on-site audit and data review.

TABLE 4. VERIFIED CORCS FOR THE WEST FRASER JOANNA MILL

Performance Metric Name / Description	Claimed Value ¹	Verified Value ²	Data Source	Reporting Period
CORC Factor	2.17 t CO ₂ / t biochar	2.16 t CO ₂ / t biochar	- LCA calculation West Fraser February 2024 v4 (with NG) (not protected).xlsx	07/01/23 to 12/31/23
Biochar Production	2773 tonne (dry)	2773 tonne (dry)	- Control Labs 10-23-2023 Analysis.pdf - Joanna Ash Totals-SEO.xlsx	

Total CORCs	6022 CORCs	5995.27 CORCs	- CORC Report Summary - Biochar_West Fraser_v2	
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¹ Claimed values are those submitted by West Fraser after completion of LCA revisions based on results of audit and extension of reporting period through 12-31-2023. Verified values are based on verification of final biochar production records for the reporting period. Output is reported in dry biochar tonnes in accordance with Puro guidance [4]
²As determined using the Puro LCA calculations workbook "LCA calculation West Fraser February 2024 v3.xlsx" [5]. Value is marked as 0 as facility does not meet the requirements set by Puro.earth.

4. AUDIT FINDINGS

4.1. SUMMARY OF AUDIT FINDINGS

35z0Solutions has reviewed and audited the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the Puro Audit Package and during the site visit and follow up:

- ☐ **Meets the requirements of the Puro General Rules V3.1 and the Annex A Biochar Methodology V3**
- ☒ **Meets the requirements of the Puro General Rules V3.1 and the Annex A Biochar Methodology V3 with minor modifications**
- ☐ **Does Not Meet the requirements of the Puro General Rules V3.1 and the Annex A Biochar Methodology V3**

A summary of specific findings associated with each requirement of the Puro Standard and Biochar Methodology and any identified issues with the audit are summarized below.

TABLE 5. AUDIT FINDINGS

Puro Standard Biochar Method. Section Ref.	Audit Verification Topic	Findings
1.1.1 5.2.1	Sustainable Feedstock	Acceptable. West Fraser obtains waste forest biomass either directly from local forest or from the co-located sawmills for all operations. Under the baseline alternative, all woody waste would typically be used as to produce biochar at the Joanna Mill, which would in turn either be used as furnace fuel or landfilled.
1.1.2 5.4.2	Biochar Use	Acceptable. West Fraser has demonstrated that the biochar will be solely sold to Scott's Hyponex for used as a soil additive. Transport emissions associated with biochar use are considered in the LCA.
1.1.3 5.2.2 5.3.3 5.4.1	Net-Negative LCA	Acceptable with modifications. West Fraser has demonstrated an appropriate basis for CORCs according to the Puro Biochar Methodology. The supplier does not currently provide sufficient numerical evidence demonstrating that plant maintenance and end-of-life emissions are below to chosen LCA cutoff. Estimates of plant operational life were also not provided.
1.1.4	Prohibition of Fossil fuel use for process heat	Acceptable with modifications. Process heat is provided by internal combustion of biomass-derived syngas. Natural Gas is only used for process startup and transition to steady operation, but it is currently not metered at the kiln level, but, rather, estimated based on total facility usage.

1.1.5	<i>Negligible methane emissions</i>	Acceptable. Emissions from the process only occur from the oxidizer exhaust (portions of the exhaust gas and the thermal oil from train C are burned in the process). The thermal oxidizer operates at conditions such that low methane emissions are anticipated. Methane emissions are measured during the annual regulatory stack test for the State of South Carolina and currently quantified at 0.11 lbs/hr.
1.1.6, 5.3.4	<i>Molar H:C Ratio < 0.7 & Biochar Qualities</i>	Acceptable. Biochar analytical results have demonstrated molar H:C ratio of 0.29. It's noted here that biochar testing was conducted on only one sample.
1.1.7	<i>Safe Environment & Biochar Handling</i>	Acceptable. West Fraser operates under an air permit issued by the South Carolina Department of Health & Environmental Control – Permit TV-1520-0078 expiration 09-30-2025. Biochar is stored in open air after quenching and cooling and is packaged off-site.
1.2.2	<i>Environmental & Social Safeguards</i>	Acceptable. West Fraser operates under the air permits discussed above and written internal safety and operational procedures. The Facility has been producing biochar for approximately 25 years in an industrial zone. West Fraser has not received any complaints from surrounding communities in recent years. The facility fits under the “retrofit” case in rule 2.1.4 of the Puro Stakeholder Engagement Requirements. This implies that additional stakeholder input is not required.
1.2.3	<i>Demonstrated Additionality</i>	Acceptable. West Fraser has submitted that: The project is not required by existing laws, regulations, or other binding obligations. Additional revenue derived from CORC sales will support West Fraser by providing needed revenue to improve the system and company financials.
1.2.4 5.3.1 5.3.2	<i>Biochar Quantification</i>	Acceptable with modifications. The final biochar and ash mix that is dispatched from the facility to Scott's is measured at the facility using a calibrated weigh scale. All product amounts included in inventory. Records of dispatch sales to Scott's Hyponex and receipts for the amount of biochar produced during this audit period were not provided. West Fraser also documents biochar and other product production in daily production logs. The LCA inputs are complete.
1.2.5	<i>Verified Production Facility standing data</i>	Acceptable. Conformance to the requirements of Section 1.2.5 of the Biochar methodology is documented and verified.
5.2.1	<i>Feedstock Sustainability Proof</i>	Acceptable. West Fraser has demonstrated that its feedstock is solely waste wood from the national forest and local sawmills (including on-site waste from the Joanna mill), which are otherwise destined for open burning or furnace feed. The company is SFI-certified. Shipping documents from the biomass transporters and weigh scale records were provided as evidence. The amount of biomass waste generated by the Joanna mill is determined by applying a 12% factor on the amount of biomass consumed at the plant. Although this does not affect the quantification of biochar production at the facility, it may affect the energy balance of the plant and therefore the LCA.
5.4.3	<i>Justification of Soil Temperature</i>	Acceptable. An appropriate soil temperature of 18.06°C in the project area was used in the LCA.
5.5.2	<i>Statement re: Double Counting</i>	Acceptable. West Fraser is aware of end-user customer practices and assures no-double counting is taking place. SEO has incorporated language in section 4.2 of its contract with Scott's Hyponex prohibiting double-counting and claiming of CORCs for use of its product.

5.5.3	<i>Marketing / Branding Restrictions on end-user</i>	Acceptable. Verifiers have found no evidence of double counting on Scott's packaging and website
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Additional details regarding audit activities, documents reviewed, and observations during the audit process are summarized in Appendix 1.

4.2. AUDIT ISSUES, REQUIREMENTS, & RECOMMENDATIONS

4.2.1. Audit Requirements

There are no major audit issues left with West Fraser's Joanna facility. All issues previously raised during the audit period have been addressed by the supplier. A summary of the issues raised by 350Solutions, and mitigating actions taken by West Fraser is presented in Appendix 1: Log of previous audit issues and responses of this report.

4.2.2. AREAS FOR FUTURE IMPROVEMENT

This section outlines requirements that West Fraser must meet in future audits to meet the requirements of the Puro Biochar Methodology. Requirements for improving the quality of data and accuracy of the LCA and CORC claims in the future include:

This procedure, and other such modifications to improve operations in the future should be noted for review in annual output audits to ensure any impacts on LCA are accounted for.

- **Natural Gas Use for Kiln Start-Up is Not Monitored**

Issue: The consumption of natural gas for kiln start-up activities is not being monitored, raising concerns about the accuracy of reported emissions. West Fraser currently estimates that natural gas consumption for kiln start-up accounts for 3.9% to 5% of the total yearly plant consumption, currently measured at 128,692MMBtu (based on estimates in the updated LCA). The emissions from this natural gas consumption was not accounted for in the original LCA submitted to Puro. A revised model accounting for these estimates of natural gas consumption was provided by Accend on 09/17/24.

Proposed Improvement: Natural Gas consumption should be monitored and reported accordingly. Emissions arising from Natural Gas consumption at the facility should be **informed by in-situ measurements rather than estimates.**

- **Purchase Records Between Scott's Hyponex and SEO Were Not Provided**

Issue: According to sections 4.4 & 5.4 of the Puro Biochar Methodology the CORC supplier must provide evidence of the end use for the biochar it produces. Due to the contractual constraints

between the supplier, SEO, and Scott's Hyponex, West Fraser does not have access to such records.

Proposed Improvement: To be eligible under the Puro Biochar Methodology, West Fraser must be able to provide delivery records to the end user, bill of landing, and/or a signed statement by the end user confirming the volume received for all deliveries following the initial audit period ending on 12-31-2023.

- **West Fraser's Analytical Plan Has Not Been Implemented Yet**

Issue: West Fraser has not yet implemented its analytical plan. The company's current plan is to follow a stringent sampling regimen to collect a statistically significant amount of data on the properties of the biochar produced. Once acceptable measurement uncertainty levels are reached, the company aims to ramp down the frequency of its testing regimen.

Proposed Improvement: It is required that changes to the sampling frequency and methodology be approved and implemented during the next audit. Verifiers should ensure that the any testing regimen is in-line with the requirements of the IBI standard and reflects any variability in the biomass produced.

4.2.3. VERIFIER RECOMMENDATIONS

This section is not related to the requirements of Puro's Biochar Methodology. It details suggested measures the CORC supplier could implement to further improve its MRV practices and surpass the requirements set by Puro.

- **Insufficient Evidence to Support that Maintenance Emissions do not Meet the Cutoff**

Issue: The supplier does not currently provide sufficient numerical evidence demonstrating that plant maintenance emissions are below to chosen LCA cutoff. Estimates of plant operational life were also not provided. Estimates of yearly parts use provided by plant staff support the assumption that the emissions are below the 1% cutoff for the facility's LCA.

Proposed Improvement: It is recommended that West Fraser monitor all maintenance parts used during upcoming audit periods and include all associated emissions in their LCA.

- **Water Use Is Not Monitored on Site**

Issue: There is no monitoring of water consumption used for quenching biochar on the project site. Although water input measurement is a requirement of the Puro methodology (section 5.3.2.), water use is expected to be minimal at the site as it uses recycled water from boiler blow off operations for quenching. However, this lack of data could lead to an underestimation of the environmental impact related to water use.

Proposed Improvement: It is recommended that water consumption be monitored in future audits. Given that the water recycling process could not be observed during this audit, monitoring will provide a clearer understanding of the project's water use and potential environmental impact.

- **Limited Information Exchange between SEO and West Fraser**

Issue: Records of deliveries to Scott's Hyponex: Sections 4.4 & 5.4 of the methodology require the CORC supplier to provide detailed evidence of the end use for the biochar it produces. Due to the contractual structure between the supplier, SEO, and Scott's Hyponex, West Fraser does not have access to such records.

Proposed Improvement: For future projects, West Fraser should strive to establish direct relationships with end users whenever possible to promote transparency across the whole biochar value chain.

- **Diesel Use and Biochar Data Not Traceable to Measurements**

Issue: Due to the nature of the relationship between SEO and West Fraser, diesel consumption and transport distances are shared as gross values for an audit period without any supporting evidence. This raises concerns about the reliability of the data used in the carbon accounting process.

Proposed Improvement: To ensure the credibility of the data, all diesel consumption and biochar production figures should be traceable to specific measurements. This involves aligning the reported data with verified transport distances and ensuring that diesel consumption data is accurately recorded and reported.

- **Biomass Waste Estimates Based on Unverified Literature Values**

Issue: The project's estimates of the biomass waste generated at the plant (12%) are based on literature values that have not been independently verified. While this is a common practice, it may introduce some uncertainty into the carbon accounting process.

Proposed Improvement: It may be beneficial to periodically review and update these estimates based on new data or research to ensure accuracy.

- **Input Biomass Moisture Content Assumed to be 50%**

Issue: The project assumes that the input biomass moisture content is consistently 50%. This value has a large impact on the process LCA as it is used to derive kiln energy production and, in turn, plant emissions. The conservativeness of this assumption during the audit period has been confirmed by measurements conducted by West Fraser.

Proposed Improvement: In the future, it is recommended to validate the input biomass moisture content as frequently as it is practical or feasible. This would more accurately reflect the facility's actual CORC production.

- **Assumption on the Ash/Biochar Ratio**

Issue: After production, biochar is mixed with electrostatic precipitator (ESP) ash. The current LCA submitted by the supplier assumes an ash to biochar ratio of 70%. The production values provided by West Fraser after the visit indicate that these values were conservative, and that the biochar-to-ash ratio was closer to 74% in practice. This metric is important as it is used to

calculate the fraction of ash delivered to Scott's Hyponex that is eligible for CORCS under the Puro methodology.

Proposed Improvement: The supplier should update the LCA with the latest Biochar to Ash ratio to reflect the latest values measured. This will ensure that any deviations from the assumed 70% ratio are captured during each audit.

- **Pyrolysis Stack Emissions**

Issue: CH₄ and N₂O emissions are currently monitored yearly. The facility does not have the capability to continuously monitor these emissions. This raises concerns about the representativeness of the emissions data measured.

Proposed Improvement: It is recommended that more frequent or continuous monitoring of the stack be implemented to verify the occurrence and effectiveness of pyrolysis gas reburn. This will provide stronger evidence to support the project's carbon removal claims. Updated pyrolyzer stack emissions should be used in the LCA when available after future regulatory driven emissions tests.

5. ELIGIBILITY FOR ONGOING ISSUANCE RIGHTS

5.1. VERIFIER'S OPINION

Puro.earth is currently transitioning to use the 4.0 version of the Puro General Rules. Although the West Fraser facility and output audit described in this report were conducted using version 3.1 of the General Rules, certain rules described in the updated version of the document (v4.0), such as the Ongoing Issuance Right, are applicable to all projects currently registered or in the process of getting registered on the registry. Specifically, Appendix A of the updated rules dictates that "the evaluation of the Ongoing Issuance Right is done in the performance verification by the 3rd party Auditor as part of the Output Audit" (A.4.1) and that "This evaluation can be done when a Production Facility has demonstrated regular industrial operation and successfully completed performance verification for the previous Monitoring Period with a minimum of 3 months of Output" (A.4.2).

Based on the age of the Joanna mill, stability of operations and production data submitted by West Fraser, 350Solutions confirms that the facility is eligible for maintain regular industrial operation and repeat the procedures for regular output monitoring. However, given that the facility has only tested the specifications of its biochar only once for verification purposes, its ability to respect the targets and timelines detailed in their proposed monitoring plan remains to be seen.

The verifier therefore recommends the issuer (Puro) to review the eligibility of facility after the first three months of issuances under the ongoing issuance rights framework, to ensure that the monitoring plan is being implemented as proposed.

5.2. ONGOING ISSUANCE ELIGIBILITY

350Solutions has reviewed and audited the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the Joanna facility:

- ☒ ***Meets the requirements for ongoing Issuance rights of the Puro General Rules V4.0 Appendix A***
- ☐ ***Meets the requirements for ongoing Issuance rights of the Puro General Rules V4.0 Appendix A with minor modifications***
- ☐ ***Does Not Meets the requirements for ongoing Issuance rights of the Puro General Rules V4.0 Appendix A***

6. REVISION HISTORY

Version	Date Issued	Noted Changes
Draft Version (v1.0)	08-29-2024	NA
Draft Version post-QA (v1.1)	09-12-2024	NA
Version 2.0	10/07/2024	Incorporated changes made to resolve non-compliance issues highlighted during the audit process. This includes: - Incorporation of estimates of Natural Gas consumption used for Kiln Startup into the LCA. - Acquisition of a letter from Scott's confirming they their awareness of the non-compliance of the testing conducted by West Fraser with IBI standards and the volume of biochar delivered during the audit period.
Version 2.1	10/15/2024	- Incorporated Accend feedback. In particular, amended text relating to end-of-life LCA emissions and facility inputs/outputs table.
Version 2.2	10/16/2024	- Incorporated Puro feedback. Added site ID number. - Included verifier's opinion on Ongoing Issuance Rights.
Version 2.3	10/23/2024	- Addresses inconsistencies between report and verifier's checklist.
Version 2.4	11/14/2024	- Amended Table 4 following Puro's comment.



AUDITOR SIGNATURES

Kyann Hadife (Lead Auditor)
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350Solutions, Inc.

Kyann Hadife

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- [2] Puro.Earth, " Biochar Methodology," Edition 2022 V3, Feb 2024. [Online]. Available: <https://puro.earth/methodologies/>.
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- [4] Puro.Earth, "Guidance for biochar suppliers on determining the dry mass of biochar produced and delivered," 2022. [Online].
- [5] Accend, "'LCA calculation West Fraser February 2024 v3'," 2024 Feb. [Online].
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APPENDIX 1: LOG OF PREVIOUS AUDIT ISSUES AND RESPONSES

This section outlines the unconformities that were addressed by West Fraser during the audit period to ensure that the Joanna facility meets the requirements of Puro's Biochar Methodology.

Findings of the LCA review, production facility audit, output verification, and on-site validation of operations included identification of nonconformities that would need to be addressed to ensure the CORCs that are compliant with the Puro Biochar Methodology can be issued.

- **Purchase Records Between Scott's Hyponex and SEO Were Not Provided**

Issue: According to sections 4.4 & 5.4 of the Puro Biochar Methodology the CORC supplier must provide evidence of the end use for the biochar it produces. Due to the contractual constraints between the supplier, SEO, and Scott's Hyponex, West Fraser does not have access to such records. A signed letter by Scott's confirming that all the biochar "contractually agreed upon" has been delivered by SEO was submitted as evidence of the biochar end use. Due to the lack of quantitative data in the declaration, it does not meet the requirements of the methodology.

Proposed Improvement: To be eligible under the Puro Biochar Methodology, West Fraser should provide either 1) an amended version of the letter by Scott's Hyponex confirming biochar delivery (see "Ash life.pdf" file), providing confirmation that the amount of biochar delivered by SEO during the 07/01/23 – 12/31/23 was received in full OR 2) an updated version of the SEO Declaration (see "Updated SEO declaration.pdf" file) with an addendum listing the total delivered in the reporting period.

If written evidence or records cannot be provided, verifiers will also accept confirmation of biochar receipt via a phone or video conferencing call with representatives from Scott's Hyponex.

Supplier Response and Mitigation: Provided updated "Ash Life # 2-ver 2(clean).docx", a letter signed by Scott's to confirm a) the volume of biochar received and b) their awareness of the non-IBI-compliant biochar testing.

- **Non-IBI-compliant biochar testing:**

Issue: As per section 1.1.7 of the Puro Biochar Methodology, suppliers must screen the biochar produced to either a) meet local limits

for polyaromatic hydrocarbons (PAH) and other compounds in agricultural uses, or b) meet the requirements of IBI, EBC, or another standard approved by Puro outlining biochar quality requirements. As option (a) does not apply in the project area, the supplier must meet IBI or EBC thresholds. West Fraser is partnering with Control Labs to ensure its biochar meets the threshold values outlined by the standard. However, the laboratory only performs IBI “Type A and C tests”. This means that the analytical results submitted as part of the audit package were missing some of the “type B tests” required by IBI - namely, tests for PAHs, dioxins/furans (PCDDs/F), Polychlorinated Biphenyls (PCBs) and the Germination Inhibition Assay. “Type B” testing for metals was conducted. According to Marianne Tikkanen's email on September 11, 2024, not conducting PAH testing is considered a deviation from the requirements.

Proposed Improvement: To demonstrate its compliance with section 1.1.7 of the methodology, West Fraser can either:

- **Test the Biochar for PAHs:** West Fraser’s must first demonstrate that the characteristics of the biochar produced by its process have been stable. This can be accomplished by providing additional testing records for biochar batches produced during the audit period (if samples are available) and showing that these characteristics are consistent with those of the biochar tested by Control Labs on 10/23/2023. Once process stability has been demonstrated, West Fraser could conduct PAH testing on one of its current batches which can be considered valid for previously produced biochar **Or,**
- **Notify Customers of the Deviation:** Inform biochar end users (Scott’s Hyponex) of the deviation from accepted standards (i.e., either not testing for PAHs or exceeding required levels). Documentation showing that customers have been informed and have approved the deviation is required. For future reporting periods, PAH testing will be necessary if the biochar is intended for use as a soil amendment or animal feed additive.

Supplier Response and Mitigation: Provided updated “Ash Life # 2-ver 2(clean).docx”, a letter signed by Scott’s to confirm a) the volume of biochar received and b) their awareness of the non-IBI-compliant biochar testing. Moreover, as per an email from Puro received on 10/7/2024, the registry considers that the [local US regulation Code 336](#) overrides the need for IBI-compliant testing as per rule 1.1.7. of the Biochar methodology edition 2022 v3 in the case of biochar applied for soil amendment. The standard does not require tests for PAHs, dioxins/furans (PCDDs/F), Polychlorinated Biphenyls (PCBs) and the Germination Inhibition Assay if the biochar is used in soil applications. Although the products made by Scott’s Hyponex are generally not used in agricultural applications, they should follow principles outlined in the code as it serves as the most relevant best practice document in the project’s jurisdiction.

- **Natural Gas Use for Kiln Start-Up is Not Monitored**

Issue: The consumption of natural gas for kiln start-up activities is not being monitored, raising concerns about the accuracy of reported emissions. West Fraser currently estimates that natural gas consumption for kiln start-up accounts for 3.9% to 5% of the total yearly plant consumption, currently measured at around 90,000 MMBtu (based on email exchange with Perry Ghuman on 07/24). According to the EPA [4], natural gas combustion alone – without upstream activities - emits around 53 kg CO₂e per MMBtu, meaning that the natural gas consumption of

the pyrolizer would represent between 1.5% and 2% of the project's total emissions during the 6-month audit period, which exceeds the 1% Cut-off criteria for the project.

Proposed Improvement: The consumption exceeds the cut-off criteria chosen by Accend, the supplier's LCA partner and should therefore be included in the project's LCA. Natural Gas consumption should be monitored and reported accordingly

- **Supplier Response and Mitigation:** The LCA, PDD Report, CORC report summary, and Mass/energy balance sheet were all amended by Accend on 09/17/24 to include estimates of natural gas consumption used for kiln startup. The values used were based on an email exchange with Perry Ghuman on 07/24.

APPENDIX 2: SUMMARY OF INSTRUMENTATION USED ON SITE

Analyte or Measured Parameter	Process Location	Sensor Type/Method	Instrument Mfg	Instrument Model	Range	Sampling or Data Recording Interval	Calibration Interval	Instrument Accuracy / Tolerance
Biochar Moisture Content	Biochar Pile	Digital Moisture Balance	CSC	SC Inc	1-100g	N/A	Each use	+/- 0.2 % moisture with a 5-gram sample
Truck Scale	Biochar Pile	Digital Truck Scale	Fairbanks	FB 2550	See truck scale certificates	N/A	See truck scale certificates	Class III/IIIL accuracy at 0–90% humidity

*Capacity programmed by West Fraser

APPENDIX 2: PURO.EARTH BIOCHAR METHODOLOGY AUDIT CHECKLIST

Production Facility and Output Audit - Biochar Methodology	
Audit ID	350PU-WF-01
Audit Inception Date	19 June 2024
Production Facility ID	West Fraser Biochar
Production Facility Location	564 Woodyard Rd, Kinards, South Carolina, 29355
Auditing Body	350Solutions, Inc.
Auditor Initials	KH
QA	LS

Guideline Ref	Requirement	Requirement Met Y/N	Verification Remarks Insert auditors' comments	Evidence Document Insert evidence used to verify requirement	Value Insert numerical value or description (if applicable)	Units Insert unit (if applicable)
Standing Data Confirmation						

1.2.5	The following standing data has been collected from Puro and checked for consistency against other evidence: - A certified trade registry extract for the CO2 Removal Supplier - CO2 Removal Supplier registering Production Facility - Evidence of the location of the Production Facility - Evidence of the Volume of Output for the full calendar year prior to registration - Evidence of the Removal Method(s) for which the plant is eligible to receive CORCs - Evidence of the date on which the Production Facility became eligible to receive CORCs - If the Production Facility has benefited from public support, evidence to show this - Documentation on Environmental and Social Safeguards imposed	Y	1. West Fraser US EWP is a limited liability company, incorporated in Delaware, with a federal tax identification number. 2. The CO2 Removal supplier is West Fraser US EWP 3. The Biochar is produced at West Fraser's Joanna facility on 564 Woodyard Road, Kinards, South Carolina, USA, 29355 4. The Joanna facility produced 2773 tonnes of biochar eligible for CORCs based on production records and logs for the facilities. This includes biochar stored in inventory as well as biochar sold for eligible applications. 5. West Fraser produces biochar via gasification of waste forest food. The biochar not used for energy purposes and sold to Hyponex, a soil additive manufacturer. - Facility uses biomass for electricity production 6. Biochar production operations became operational several years ago, as a bi-product of waste biomass gasification on-site. An outcome of this gasification on process is the continual production of biochar which has accumulated onsite for many years. Evolving permitting requirements have urged West Fraser to look at a more efficient approach to managing this 'waste' byproduct, of the OSB's production on process. During the period July-Dec 2023, West Fraser delivered 2,773 dry tonnes of biochar to be used in soil products which will be	1. Tab 04.09 Certificate of Merger 2. West Fraser EWP LLC certificate of formation filed with DE SOS 11-27-2023 3. BB_Stakeholder Engagement Report_ver2 4. Final PDD West Fraser- Joanna Facility. 3.5.2024 5. Ash moisture readings.docx 6. JOA_Terms and Conditions.pdf 7. NSF CoC C0750561-CC1 (1).pdf 8. NSF Fiber Sourcing C0750561-FS2 (2).pdf 9. Residual and log purchase contracts at West Fraser.docx 10. Sustainable Forestry Initiative (SFI) Fibre Sourcing - all wood procurement WF US mill.pdf 11. JOA_Environmental Evaluation Report_v1_signed.pdf		
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			counting towards the CORC claims as they were produced less than 18 months before the verification period. 7. West Fraser has not received public funds to support operation of its facility 8. West Fraser US EWP operates under a Title V Air Operating Permit (TV-1520-0078) issued by the South Carolina Department of Health and Environmental Control. The facility has a stormwater permit.			
Evidence Confirmation						

5	All necessary evidence has been provided to the auditor by the Production facility as per Section 5 of the Biochar Guidelines and has been used to complete the compliance checklist. - Proof of sustainability of raw material for forest biomass (FSC, SFI, PEFC, other certifications) - Proof of sustainability of raw material for waste biomass - LCA data for biomass and biochar production, supply and use, including climate change impact and the contribution of each life cycle stages. - Proof of product quality: laboratory analysis of total organic carbon content, hydrogen content and H/Corg - Proof of production volume: documentation for the whole period and methodology applied to calculate the dry mass of biochar produced. - For mobile units or carbonizer operator: proof of load cell measurement of the biochar for the whole period, and water input measurement. - Proof of end use of biochar: offtake agreement, shipment, and other records indicating the intended use of biochar. - Justification on the soil temperature selected for the calculation of the biochar sequestration. - Proof of sales - Proof of no double counting/C positive marketing	Y	1. Raw material is sourced from three sources: Purchased utility-size waste wood chips and grindings, that come from the tops and limbs of harvested trees from SFI-audited landscapes. Species include the Southern Yellow pine family - Southern Yellow, Longleaf, Slash, Shortleaf, and Loblolly pine- as well as a mix of so or hardwoods inc. sweet gum hardwood, and various species of oak (Red and White oak). The chipping and grinding takes place in the forest. SFI compliance certificates have been provided. Purchase agreements mandate supplier compliance with State Best Management Practices (BMP's) and applicable Sustainable Forest Initiative (SFI) Principles and Objectives. Purchased waste wood that includes bark, sawdust, and hog fuel from sister mills in the region and from forest thinning's to support forest health. This arrives to the Joanna facility as moist chips that have already been processed. Waste bark from the peeling of the logs in the production of the OSB boards at the Joanna facility. 2. West Fraser has also provided an NSF certificate of conformity for chain-of-custody management. No evidence of historical biomass use/fate for any of the sources above. 3. LCA is currently incomplete.	1. LCA calculation spreadsheet (file: LCA calculation West Fraser February 2024 v3.xlsx) 2. NSF CoC C0750561-CC1 (1).pdf 3. NSF Fiber Sourcing C0750561-FS2 (2).pdf 4. Sustainable Forestry Initiative (SFI) Fibre Sourcing - all wood procurement WF US mill.pdf 5. Control Labs 10-23-2023 Analysis.pdf 6. 7-28-23 Inbound truck scales certification,calibration tests (1).pdf 7. 7-28-23 Outbound (truck scales certification,calibration tests 1).pdf 8. 9-27-23 Inbound truck scales certification,calibration tests (1).pdf 9. 9-27-23 Outbound truck scales certification,calibration tests (1).pdf & Joanna Ash Totals-SEO/xlsx 10. Signed Declaration from Scott's.pdf 11. Joanna Ash Totals-SEO.xlsx 12. Soil Temperature Selection Methods.docx 13. July 2023 Wood Usage.pdf 14. August 2023 Wood Usage.pdf 15. Sept 2023 Wood Usage.pdf 16. Mill Wood Usage Oct 2023.pdf 17. Mill Wood Usage Nov 2023.pdf 18. Mill Wood Usage Dec 2023.pdf		
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			(see items 1.1.3 and 3.2) 4. Lab test completed by Control Labs for biochar ult/prox analysis (and HC ratio). West Fraser also analyzes product for moisture content using moisture balance/meter for char quality indicator. Moisture content was also measured independently for a batch produced between 10/24/2023 and 10/31/2023 (Chris Cagle Email). 5. Production logs are provided documenting all biochar production, including weighed biochar amount and moisture content of biochar using appropriate methods. Direct scale readings are not provided. 6. Biochar is measured on calibrated weigh scale at the Joanna facility. Joanna scales calibrated at pre-scheduled intervals by certified calibration service (ANAB accredited for ISO 17025). Biomass is also weighed on weigh scale at supplier facilities (not current calibration). Water usage is not currently monitored and no estimates are provided. Direct scale readings are not provided. 7. Biochar use is not documented. Hand application is expected given the nature of the soil amendment product sold by Scott's Hyponex. No invoices or records of sales were provided by West Fraser, but a		
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			declaration confirming biomass receipt has been signed. 8. Soil temp used is average soil temp and meets average for South Carolina/Tennessee area where biochar is sold and used 9. West Fraser has not supplied invoices for any sales of biochar for CORCs. Inventory records (weigh scale measurements) are provided for all biochar produced and shipped from the facility. 10. West Fraser will add language to quotes stating C positive/C credit claims cannot be marketed by end users of biochar. No evidence of any such double counting or marketing observed, but no requirements currently in place.			
Eligibility Checklist						

1.1.1	Biochar is used in applications other than energy.	y	(Criterion Met) - All applications claimed for CORCs are indicated records as soil amendments. This was confirmed by a letter from Scott's Hyponex.	Ash Life.pdf		
1.1.2	Biochar is produced from sustainable forest or waste biomass raw materials (consult list of raw materials).	Y	The feedstock is mainly from forest thinning's, the tops and limbs from sustainably managed forests and waste fiber from processing industries (forest, sawmills, trees from SFI-audited landscapes).	On site observation, West Fraser SFI Certification (Available at https://www.westfraser.com/sustainability/forest-certification). July 2023 Wood Usage.pdf August 2023 Wood Usage.pdf Sept 2023 Wood Usage.pdf Mill Wood Usage Oct 2023.pdf Mill Wood Usage Nov 2023.pdf Mill Wood Usage Dec 2023.pdf		
1.1.3	LCA shows: - carbon footprint of the biomass production and supply - emissions from the biochar production process - carbon footprint of the biochar end use - cradle to grave	y	See 5 (item 2) above. LCA is currently complete and accurate.	LCA calculation spreadsheet (file: LCA calculation West Fraser February 2024 v3.xlsx),		
1.1.4	Pyrolysis reactor input fuel for heating is not a fossil fuel. Unless only used for ignition/pre heating or in a mobile unit and the emissions are fully included in the LCA. The use of waste heat from other industrial processes (eg. Biodigesters, cement production) is permitted.	Y	The plant is 100% biomass-fired. Natural gas is used in the secondary combustor on chamber as a fallback measure to produce steam. Natural Gas is only used during the Kiln start up procedure, but its usage is not directly associated with biochar production.	Not measured		
1.1.5	Pyrolysis gases are combusted or recovered. Bio-oil and pyrolysis gases can be stored for later use as renewable energy or materials.	Y	The pyrolysis gas is combusted with energy recovery.	Not Observed. Based on mass balance and stack testing results.		
1.1.6	The molar H/Corg ratio is less than 0.7.	Y	Tested 0.29. Only one sample.	Control Labs 10-23-2023 Analysis.pdf		

1.1.7	Evidence of safe handling and transport is provided and adequate for the production facility.	Y	The West Fraser SDS and OSHA documents identify and propose mitigation strategies for all the potential H&S risks associated with biochar handling and transport. Moisture contents are measured by Control Labs (External Laboratory)	Joanna Biochar OSHA-WHMIS-GHS SDS 2024-03-01 & Joanna_West Fraser Ash SDS.pdf		
Production Facility Checklist (Desktop and Verbal Confirmation)						
1.2.1	Evidence of Production Facility eligibility under the general rules of Puro Standard.	Y	West Fraser Meets all Puro eligibility criteria (see above items)	See above items		
1.2.2	The Production Facility demonstrate Environmental and Social Safeguards.	Y	See above (1.2.5). Air permits provided are expired. Stormwater permits are not provided.	See 1.2.5		
1.2.3	CO2 Removal Supplier shall be able to demonstrate additionality, meaning that the project must convincingly demonstrate that the CO2 removals are a result of carbon finance. Even with substantial non-carbon finance support, projects can be additional if investment is required, risk is present, and/or human capital must be developed. To demonstrate additionality, CO2 removal Supplier must provide full project financials and counterfactual analysis based on Baselines that shall be project-specific, conservative and periodically updated. Suppliers must also show that the project is not required by existing laws, regulations, or other binding obligations.	Y	CORCs are utilized to support business operations in addition to revenues from the generation of electricity, and sale of biochar and kiln ash. Addition of CORC revenues will improve company financials (currently operating at a loss), and enable expansion of production. Although recent regulation changes have mandated a shift in the biochar management practices at the Joanna facility, biochar use for soil amendment is not required by law.	Puro additionality - West Fraser - Joanna June 2024.pdf & Joanna Financial Additionality.xlsx		

1.2.4	The Production Facility's documentation system is accurate and reliable	Partially	Facility utilizes daily production and operating logs, biomass weight log, Biochar analysis logs (for only one batch is provided). Production logs are based on weigh scales at (details to be discussed during visit). All claimed CORCs based on weights of sold/shipped products. Recommend improving documenting of diesel fuel use and natural gas. Also improving document/monitor water use. No records of product delivery were provided.	N/A		
	The quantity of the biochar produced and sold is quantified and documented in a reliable manner	Y	Logs for biochar production are provided for the whole accounting period. No sales data or records of delivery to Scott's have been provided. Signed letter by Scott's confirms delivery of claimed biochar volume.	Ash life.pdf & Updated SEO declaration.pdf & CORC Report Summary - Biochar_West Fraser.xlsx & Ash Life # 2-ver 2(clean).docx	2773	tonne
	Relevant meters are in place and they are calibrated;	Partially	Weigh Scale meters are calibrated (aside from CART15503B which is now due for calibration - not due at the time of the application however) by qualified calibration entity. All biochar products are weighed on this scale at the Joanna facility. Meter calibration for moisture meter/balance should be provided. Water metering is not currently completed and no estimates are provided. Stack emissions are monitored yearly for permitting reasons.	On site observation & Joanna Biochar sampling and dry mass calculation protocol_ver2.doc & Control Labs 10-23-2023 Analysis.pdf & 7-28-23 Inbound truck scales certification, calibration tests (1).pdf & 7-28-23 Outbound (truck scales certification, calibration tests 1).pdf & 9-27-23 Inbound truck scales certification, calibration tests (1).pdf & 9-27-23 Outbound truck scales certification, calibration tests (1).pdf & Joanna Ash Totals-SEO/xlsx		

	The emissions from the cultivating, harvesting and transporting of the biomass are estimated and calculated in a reliable manner	y	Biomass transport is included in LCA. Harvesting and cultivating are not included, as this is waste biomass bound for landfill otherwise. No special handling for biomass required. On-site transport and handling is considered non-material.	LCA calculation West Fraser February 2024 v3		
	The energy use of the Production Facility can be quantified and the emissions from the process calculated	Partially	Energy use on site is net-metered for electricity. All energy used for site operations (electrical) is provided by the gensets prior to metering to the grid. Natural gas consumption (used for kiln start up) records are not provided.	LCA calculation West Fraser February 2024 v3		
	The auditor goes through the Quantification of CO2 Removal requirements with the CO2 Removal Supplier, so that the Supplier is able to calculate the CO2 Removal independently in its Output Report	y	Records of inbound biomass and outbound biomass are kept as invoices and measurements. No records of sawmill residues and invoices from Scott's Hyponex have been provided.	Control Labs 10-23-2023 Analysis.pdf & 7-28-23 Inbound truck scales certification,calibration tests (1).pdf & 7-28-23 Outbound (truck scales certification,calibration tests 1).pdf & 9-27-23 Inbound truck scales certification,calibration tests (1).pdf & 9-27-23 Outbound truck scales certification,calibration tests (1).pdf & Joanna Ash Totals-SEO/xlsx		
3.1	The CO2 Removal Supplier shall provide a life cycle assessment (LCA) for biochar activity including disaggregated information on the emissions arising at different stages. The system boundary is set cradle-to-grave and shall include emissions from production and supply of the biomass, from biomass conversion to biochar, and from biochar distribution and use.	y	Boundary currently set as Cradle-to-grave. Plant construction and end of use are not accounted for as no changes were made to that plant to support CDR operations. All emissions are accounted for within system boundary after LCA modifications were completed. Biomass handling/supply and biochar end use applications do not require additional handling beyond normal existing operations, so no Lca components associated with either are included (except transportation to sites).	On site observation, LCA calculation West Fraser February 2024 v3.xlsx		
3.2	Life cycle assessment (LCA) shall follow ISO standard, WRI GHG protocol or similar method.	y	Fulfills the requirements for a cradle-to-grave analysis. See section 4.4.	LCA calculation West Fraser February 2024 v3.xlsx		

3.3	The default baseline emission scenario for the project activity feedstock is zero, which is a conservative assumption since it is not taking into account methane emissions derived from decay of manure or combustion of waste biomass. If a non-zero baseline presented, needs to be accepted by Puro.earth	Y	Not applicable.	NA		
Calculation Checklist						
4.2	Qbiochar = Quantity of biochar produced and sold to end user. (dry char)	Y	Confirmed	LCA calculation West Fraser February 2024 v3.xlsx and	2773.00	tonne (used)
	FpTHTs = $c + m \times H / C_{org}$	Y	Confirmed	LCA calculation West Fraser February 2024 v3.xlsx	83.0%	%
	C Biochar = carbon content of biochar	Y	Confirmed	LCA calculation West Fraser February 2024 v3.xlsx	76.5%	%
	Estored = biochar carbon storage = $Q_{biochar} \times C_{biochar_{org}} \times F_{pTHTs} \times 44/12$	Y	Confirmed	LCA calculation West Fraser February 2024 v3.xlsx	2329	kg CO2e/tonne biochar (dry) used
4.3	Ebiomass = LCA emissions of production and supply of biomass	Y	Confirmed	LCA calculation West Fraser February 2024 v3.xlsx	21	kg CO2e/tonne biochar (dry) used
4.4	Eproduction = LCA emissions from biochar manufacturing	Y	No. Eproduction does not include plant construction and EOL emissions as EOL and construction do not meet the LCA Cutoff Criteria set by Accend.	LCA calculation West Fraser February 2024 v3.xlsx	119	kg CO2e/tonne biochar (dry) used
4.5	Euse = LCA emissions of the use of biochar, including distribution up to the point of final use	Y	Confirmed	LCA calculation West Fraser February 2024 v3.xlsx & Joanna Ash Totals-SEO.xlsx	27	kg CO2e/tonne biochar (dry) used
4.1	CORCs = Estored - Ebiomass - Eproduction - Euse	Y	Confirmed	LCA calculation West Fraser February 2024 v3.xlsx & CORC Report Summary - Biochar_West Fraser.xlsx	5995	Tonnes of CO2e
	Quantity of CORCs (in evidence).	Y	Confirmed	CORC Report Summary - Biochar_West Fraser.xlsx	5995	CORC
	Confirm consistency.				Y	

APPENDIX 3: 350SOLUTIONS/WEST FRASER POST-VISIT REQUEST FOR INFORMATION

Document instructions: Kindly provide answers to the questions in this file in the format you deem appropriate.

Items requested after the site visit

- Kindly provide the feedback received from Puro regarding your sampling plan.
- Kindly provide the sampling/monitoring plan in its latest form (to be discussed with Puro)
- Kindly provide the updated NPV calculations file once it is ready.
- If possible, kindly perform more moisture content measurements on the biomass used in the energy system to support the 50% biomass moisture content assumption.
- Kindly provide a reference/justification (could be from your own records) for the 12% bark per log assumption used at the facility.
- Kindly provide operational logs for the Pyrolysis gas reburn (if available).
- Kindly provide a reference for the \$1200/t transport + tipping fee assumption in the additionality calculations.
- Kindly provide any information possible (if any becomes available) on natural gas and water consumption in the process.

Items that could potentially be requested following 350Solutions' call with Accend.

- Biochar handling equipment (conveyor belts, on-site trucks, ...) diesel consumption records and embodied emissions.
- Maintenance/repair records and schedules for the energy system (to determine the plant's embodied emissions)
- Plant assumed operating life (based on conversation with Puro and Accend).

APPENDIX 4: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications

Company Name:	West Fraser
Date:	7/11/2024
Verifier Name:	Kyann Hadife
Company Name (where applicable):	350Solutions
Verifier Contact Information:	kyann@350solutions.com , (332)265-5282
Verifier Address:	1053 E. Whitaker Mill Rd. Suite 115, Raleigh, NC 27604
Verifier Scope of Activities:	Production facility and output audit through review of facility standing data, on-site observations and meeting, review of key technology components, operational data, and documentation.

Verifier Qualifications	Criteria Met?	Evidence / Notes
(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.)		
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	Yes	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Puro.Earth General Standard. Note that verifications completed for Puro.Earth are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanently stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to	Yes	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	Yes	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	Yes	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	Yes	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	Yes	
Verifier is a credible independent 3 rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Puro.Earth CO2Cs;	Yes	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Puro.Earth for verification, or represent the parties engaged in those activities;	Yes	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items	Yes	

William Chatterton
350Solutions, Verification Program Manager

EDUCATION

B.S. Environmental Science, SUNY at Plattsburgh, 1982

Certified Measurement and Verification Professional (CMVP), 2019

Professional Experience

William Chatterton is an Environmental Scientist with 28 years' experience in technology evaluation and demonstration, project management, air pollution monitoring, testing, and regulation. He serves as Program Manager at 350Solutions and manages projects and programs for commercial and government clients. Previously the past 20 years at Southern Research, Mr. Chatterton has managed, and supported programs designed to integrate, demonstrate, and evaluate technology performance in the advanced energy field. Technology demonstrations and evaluations that he has been involved with include technologies designed to promote sustainable energy sources, increase energy use and efficiency, mitigate GHG and other emissions, and in most cases provide other social and economic benefits to potential users. Mr. Chatterton has been heavily involved in the evaluation of numerous emerging energy technologies, distributed generation technologies, and technologies relevant to transportation and oil and gas markets. Mr. Chatterton's roles in support of these projects has included program and project management from administrative and technical perspectives, lead or technical support on test plan development, method development and validation, design and implementation of field-testing activities, data evaluation and presentation, and reporting of results. He has managed numerous projects for both commercial and government clients.

350Solutions: 08-2019 – Present

Verification Program Manager: As Verification Program Manager, Mr. Chatterton manages and executes technology performance demonstrations and verifications of emerging energy (efficiency and green building) and transportation technologies, primarily for U.S. governmental agencies, energy research associations, and state energy agencies. These performance evaluations generally involve evaluation of commercial feasibility, economic impacts (installation, operating, and capital costs, simple payback, and return on investment), environmental impacts (primarily greenhouse gas and criteria pollutant emission reductions), and technology performance. He also manages and monitors 350Solutions' quality management programs and ISO accreditations.

Southern Research Institute: 1999 - 2019

Program Manager, Energy & Environment Technologies: As Program Manager, Mr. Chatterton has managed and executed several technology performance demonstrations and verifications of emerging energy (efficiency and green building) and transportation technologies, primarily for U.S. governmental agencies, energy research associations, and state energy agencies. Mr. Chatterton also has direct experience with management and execution of projects under DOE and DoD grants and contracts. He has recently managed activities on three large DoD projects including Demonstration of a Solar Thermal Combined Heating, Cooling and Hot Water System Utilizing an Adsorption Chiller for DoD Installations, Demonstration and Verification of the Performance of Microturbine Power Generation Systems Utilizing Renewable Fuels, and the Electric Power with Small Scale Organic Rankine Cycle (ORC) Engine/Generator Technology demonstration.

Kyann Hadife
350Solutions, Carbon Removal Verification Engineer

EDUCATION

Columbia University - MS Carbon Management (4.03/4.0 GPA)

University College London - BS Civil & Environmental Engineering (4.0/4.0 GPA)

SUMMARY

Kyann is an MS in Carbon Management graduate from Columbia University with experience in LCA/TEA consulting, laboratory analytical work, and engineering design. Prior to Columbia, he earned a bachelor's degree in civil and environmental Engineering from University College London. At UCL and Columbia Kyann has researched anaerobic digestion, Direct Air Capture and CO₂ mineralization, developing prototypes, SOPs, and data analysis models. During his master's, Kyann also consulted for companies across hybrid and engineered CDR pathways including BiCRS, OAE, and Mineralization, on developing LCA/TEAs and site-selection tools, conducting market research, and writing applications for grants from the DOE, XPRIZE foundation, Remove and others. Most recently, Kyann has spearheaded Carbon Lockdown's Puro.earth verification for their FOAK terrestrial biomass burial project. Before being a full-time team member, Kyann worked as a research intern at 350Solutions in the summer of 2023.

RESEARCH AND PROFESSIONAL EXPERIENCE:

Mar 2024 – Present: 350Solutions, Carbon Removal Verification Engineer

At 350Solutions, Kyann has Verified CDR technologies on behalf of the XPRIZE Carbon Removal, registries challenge across pathways, including DAC, DOC, and mineralization. During his time at the company, he has also verified monitoring approaches (ISO14034) and validated LCA models (ISO14040) for carbon removal approaches.

Nov 2023 – Mar 2024: Carbon Lockdown Project, MRV Consultant

Kyann has spearheaded certification efforts for Carbon Lockdown's 5kt/y Potomac site under Puro.earth's Terrestrial Biomass Burial Methodology, co-developing the company's LCA, TEA, EIA, and MRV plan. He has also evaluated the eligibility of the project biomass under various sustainable biomass sourcing standards. As part of this role, Kyann participated in webinars and working groups organized by the DOE and national laboratories aiming to develop a unified LCA framework for terrestrial biomass burial.

Sep 2023 – Mar 2024: Brineworks, LCA/TEA Consultant

During his time at Brineworks, Kyann has conducted mass, energy, and carbon balances for 6+ carbon-negative brine treatment and material recovery pathways. He used LCAs and TEAs to inform process optimization efforts and achieve net negative process emissions while maintaining competitive brine treatment prices. He has also investigated the additionality and economics of a variety of CDR pathways unlocked by electrochemical brine treatment.

Sep 2023 – Dec 2023: Master's Thesis – Columbia University & Octavia Carbon

Conducted a cradle-to-gate LCA/TEA on project Hummingbird, the global south's first DAC + Mineralization plant. As part of the study, Kyann leveraged Aspen HYSYS and Adsorption models, along with statistical methods, to predict plant performance at scale based on values obtained from lab-scale testing. As part of his academic coursework at UCL and Columbia, Kyann has also conducted LCAs on various processes and products including Kenaf fiber insulation board, Steam Methane Reforming with pressure swing adsorption, and Biogenic/Algal CO₂ conversion.



CERTIFICATE OF ACCREDITATION

The ANSI National Accreditation Board

Hereby attests that

350Solutions, Inc.
1053 E. Whitaker Mill Rd., Suite 115
Raleigh, NC 27604

Fulfills the requirements of

ISO/IEC 17020:2012

and

**ISO 14034:2016, Environmental Management - Environmental
Technology Verification (ETV)**

In the field of

INSPECTION

This certificate is valid only when accompanied by a current scope of accreditation document.

The current scope of accreditation can be verified at www.anab.org.



R. Douglas Leonard Jr., VP, PILR SBU

Expiry Date: 25 September 2024

Certificate Number: AI-2618



An inspection body's fulfilment of the requirements of ISO/IEC 17020:2012 means the inspection body meets both the technical competence requirements and management system requirements that are necessary for it to consistently deliver technically valid inspection results (refer to joint ISO-ILAC-IAF Communiqué dated Sept 2013).



SCOPE OF ACCREDITATION TO ISO/IEC 17020:2012

and

ISO 14034:2016, Environmental Management - Environmental Technology Verification (ETV)

350Solutions, Inc.
1053 E. Whitaker Mill Rd., Suite 115
Raleigh, NC 27604
Tim Hansen tim@350Solutions.com
(919) 675-6432

INSPECTION TYPE A (THIRD-PARTY) BODY

Valid to: **September 25, 2024**

Certificate Number: **AI-2618**

General

Products Categories	Range	Stage	Methods and Procedures
Energy Technologies (ET):	Performance and Environmental impact as it relates to design, materials, equipment, installation and operations.	Operating	QSP-350-223-02 - SOP ISO 14034 ETV
Cleaner Production and Processes (CPP):	Performance and Environmental impact as it relates to design, materials, equipment, installation and operations.	Operating	QSP 350-223-02 - SOP ISO 14034 ETV
Air pollution monitoring and abatement (APP):	Performance and Environmental impact as it relates to design, materials, equipment, installation and operations.	Operating	QSP 350-223-02 - SOP ISO 14034 ETV
Water monitoring and treatment (WMT):	Performance and Environmental impact as it relates to design, materials, equipment, installation and operations.	Operating	QSP 350-223-02 - SOP ISO 14034 ETV

Note:

1. This scope is formatted as part of a single document including Certificate of Accreditation No. AI-2618.



R. Douglas Leonard Jr., VP, PILR SBU

Version 005 Issued: August 09, 2022

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